

Abhishek Sen

Roll No.: EC22B1098

Bachelor of Technology

Indian Institute of Information Technology Design and Manufacturing,
Kancheepuram CGPA: 8.38 [2022–26]

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🔗 GitHub Profile

🌐 LinkedIn Profile

🌐 Portfolio Website

ABOUT ME

An enthusiastic engineering graduate specializing in Robotics , Embedded system and Electronic hardware. Proven ability to develop and deploy robust, real-time robotics solutions, ranging from Edge AI systems (Jetson Orin Nano) to complex SLAM-based autonomous navigation. Highly motivated to apply advanced Robot Framework ROS2 , MAVROS on robotic arm and mobile robot to solve challenging, real-world perception problems.

EXPERIENCE

•Project Assistant at IIT Madras

Guidance, Navigation and Control Lab, IIT Madras
[Research Paper Link]

October 2025 – Present

- Implemented a proprietary path planning and navigation system using the **Generalized Shape Expansion (GSE)** algorithm within a Gazebo simulation environment and real time in hardware using ROS2 and MAVROS.
- Deployed the navigation stack on an **NVIDIA Jetson Orin Nano** companion computer, utilizing a Livox 4D LiDAR for high-fidelity real-time data collection.
- Demonstrated that the GSE algorithm significantly outperformed traditional sampling-based methods (such as RRT*) in both speed and accuracy regarding obstacle avoidance.
- Implemented sensor fusion pipelines using **Kalman and Particle Filters** to integrate IMU and 4D LiDAR data, ensuring precise state estimation and odometry.
- **Technologies:** ROS2 Humble, SLAM, Gazebo, Online GSE Algorithm, ArduPilot Firmware, Sensor Fusion.

•hardware Research and development Intern

32Mins Digital Consultancy Services, IIT Madras Research Park

May 2025 – Sep 2025

- Worked on an **AI-driven learning partner device** to enhance student engagement during online classes.
- Worked on texas instrument board TDA4VM hardware device for dual camera callibration
- Also wrote camera custom driver and implemenated in docker container
- Improved user interaction and feedback loop to personalize learning experiences using AI models.
- **Technologies:** Embedded system, Edge AI, TDA4VM, Docker , Image processing

•Project Intern

Autonomous System Laboratory

May 2024 – July 2024

- Developed an autonomous robot capable of navigating complex environments using LIDAR, SLAM, and Computer Vision simulation as well as in real time implemenation
- Equipped the robot with real-time obstacle detection and avoidance for safe and efficient navigation.
- Implemented the GMapping algorithm to generate precise maps of the environment.
- **Technologies:** Computer Vision, ROS2, SLAM, GMapping [View LOR Here]

PERSONAL PROJECTS

- **Autonomous Docking and Visual Servoing Robot (ROS 2)** 🔗 [GitHub Link]
- Developed a ROS 2 package for autonomous docking that utilizes computer vision (HSV segmentation) to detect and track specific targets in a Gazebo environment.
- Implemented sensor fusion by integrating monocular camera data (for bearing) with LiDAR LaserScan data (for distance) to achieve precise target localization.
- Implemented a proportional controller to align and drive the robot toward targets while simultaneously executing obstacle avoidance logic.
- Designed a visualization pipeline publishing dynamic TF frames and markers to RViz2 for real-time tracking of detected objects.
- *Technologies:* ROS 2 Humble, OpenCV, Sensor Fusion, TF2, Gazebo, Python, RViz2
- **Robust Navigation and Trajectory Control (ROS 2)** 🔗 [GitHub Link]

- Developed a custom lightweight navigation stack for differential drive robots (TurtleBot3), bypassing heavy global planners to achieve high-frequency (20Hz) control in unstructured environments.
 - Implemented B-Spline Trajectory Smoothing to convert sparse waypoints into continuous paths with C^2 **continuity**, ensuring fluid kinematic profiles.
 - Implemented Pure Pursuit Controller optimized with a **Sequential Index Search (Sliding Window)** to robustly track looping paths without backtracking errors with obstacle avoidance.
 - *Technologies: ROS 2 Humble, Python (NumPy, SciPy), Gazebo, Control Theory, RViz*
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- **Autonomous Navigation of drone for Martian Terrain Analysis, Drone which can fly on mars**
 - Developed an Autonomous Drone for the ISRO Robotics Challenge, achieving **GPS-less navigation** using sophisticated vision processing.
 - Integrated **Simultaneous Localization and Mapping (SLAM)** using the **Intel RealSense Depth Camera** for real-time environmental mapping.
 - Implemented **YOLOv8** for robust object/boundary detection. Utilized ROS2/MAVROS to convert vision data to real-time local position estimates.
 - *Technologies: Deep Learning, ROS2, MAVROS, SLAM, YOLOv8, RANSAC*
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- **Autonomous Drone for Real-Time Surveillance and Object Tracking**
 - Built an autonomous drone system leveraging **YOLOv8** for object detection and tracking during autonomous flight missions drone is fully designed and customise with full software stack.
 - Demonstrated robust real-time perception capabilities vital for dynamic surveillance environments.
 - *Technologies: YOLOv8, Machine Learning, ROS2, Computer Vision*

POSITION AND RESPONSIBILITIES

•Team Captain

July 2023-Jan 2026

Aerobatics Team Astra

- In-depth understanding Of Flight controller and Propulsion system of drone.
- Developed Skills like PCB design ,Team management ,Arduino IDE,wireless communication using mavlink protocol
- Strong Knowledge About drones and interested in arial robotics ROS2 ,AI/ML
- Experience with hardware companion computers, including Jetson Orin Nano and Raspberry Pi, for enhanced robotic functionality

•Co-Curricular Activities

2023- Jan 2025

- NCC Cadet(National cadet corps)
- dramatics Cultural club coordinator

TECHNICAL SKILLS AND INTERESTS

DeepLearning/Computer Vision: YOLOv8, ResNet-18, SLAM (GMapping), PyTorch, TensorFlow, Keras, OpenCV, dlib, Real-Time Image Classification

Languages: C/C++,OOPS, Python, Javascript, HTML+CSS

Libraries : TensorFlow, PyTorch, Keras, Scikit-Learn, C++, Python Libraries like dlib,cvzone,face qrecognition,Pandas,numpy,matplotlib

Web Dev Tools: VScode, Github,Jupyter notebook

Hardware: Resberipy,Arduino ide,ESP32,GPS,GSM module .

PCB Design: Altium software , EasyEDA

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming,Computer Vision .

Areas of Interest: Robotics,Computer Vision , AI/ML .

Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability

ACHIEVEMENTS

- Top Finalist, ISRO Robotics Challenge – Appreciated for Autonomous GPS-less Drone at National Space Day, Delhi by ISRO Chairmen and astronauts.**
- Secured AIR 8 at SAE Aerothon 2025 with Team Astra for Detailed Design Report and Presentation Round.**
- Winner, SAE Aerothon Challenger Round; Achieved AIR 21 in Report Documentation Round.**
- One of four teams at IIT Roorkee to successfully complete both manual and autonomous drone missions.**